

mycv

Mathematical Foundations

A Pure-NumPy Computer Vision Library

Abodunrin Charles Toluwanimi

Department of Electrical and Electronic Engineering
Obafemi Awolowo University, Ile-Ife

An updated mathematical and implementation reference for mycv v4.2.1. This edition extends the earlier v2.0.0 and v3.0.0 notes with tracking, calibration, region geometry, neural networks, and the bundled shape classifier.

Version 4.2.1 | September 2026

Contents and conventions

This reference follows the source tree and public API of mycv v4.2.1. The core image algorithms use NumPy and do not require OpenCV, SciPy, or scikit-image. Optional display, image I/O, and streaming features use separate packages.

Image domain: $\Omega = \{0, \dots, H-1\} \times \{0, \dots, W-1\}$
 Coordinates: $(x, y) = (\text{column}, \text{row})$; array indexing is `image[y, x]`.
 RGB shape: $(H, W, 3)$ Grayscale shape: (H, W)
 Mask foreground: $M[y, x] > 0$
 Box: $(y1, x1, y2, x2)$, exclusive bottom-right; `crop = image[y1:y2, x1:x2]`

Module map

core/color: pointwise maps, luminance, histogram and HSV. filters/morphology: discrete operators, set morphology and connected components. geometry/calibration: interpolation, projective maps and homography estimation. features/detection: local moments, shape descriptors, template matching and overlap. tracking: segmentation, state estimation and data association. nn/shape_cnn: reverse-mode differentiation and learned shape classification.

Coordinate order and endpoint conventions matter when data moves between modules. In particular, boxes use row-first order and exclusive end coordinates.

1. Point operations and colour

RGB to grayscale

For RGB components in $[0, 255]$, the library applies the BT.601-style luminance map. It computes in floating point, clips, rounds, and returns uint8. This is a luma-like grayscale conversion, not a camera-profile-aware colourimetric transform.

$$Y = 0.299 R + 0.587 G + 0.114 B$$

Threshold

A global inclusive threshold partitions grayscale values into foreground and background. A single threshold may be inadequate under changing illumination.

$$T_{\text{tau}}(I)[y, x] = 255 \text{ if } I[y, x] \geq \text{tau, otherwise } 0$$

Histogram equalization

Let $h(k)$ count pixels at intensity k and $N=H W$. The empirical probability mass and cumulative distribution define an intensity lookup table.

$$\begin{aligned} p(k) &= h(k)/N & F(k) &= \sum_{j=0..k} p(j) \\ L(k) &= \text{round}(255 F(k)) & I_{\text{eq}}[y, x] &= L(I[y, x]) \end{aligned}$$

RGB to HSV and masks

Normalize RGB to $[0, 1]$. Let $V=\max(R, G, B)$, $m=\min(R, G, B)$, and $\Delta=V-m$. Saturation is Δ/V for $V>0$ and zero otherwise. Hue is selected piecewise from the maximum channel and lies in $[0, 360)$; saturation and value lie in $[0, 1]$. Achromatic pixels use hue zero by convention.

$$\begin{aligned} S &= \Delta/V \text{ when } V>0, \text{ else } 0 \\ H &= 60 * ((G-B)/\Delta \bmod 6) \text{ if } V=R \end{aligned}$$

```
H = 60*((B-R)/Delta + 2)    if V=G
H = 60*((R-G)/Delta + 4)    if V=B
```

Colour masks use inclusive HSV range checks. The hue-wrap helper handles intervals crossing 0/360. Adaptive masking uses saturation and value thresholds for near-achromatic samples, where hue is numerically unstable.

2. Filters and morphology

Discrete convolution

The implementation performs true convolution by reflecting the kernel along both axes. Same mode uses zero padding and preserves image shape; valid mode uses no padding. A strided patch view and einsum contract each patch with the kernel.

$$(I*K)[y,x] = \sum(i,j) I[y-i,x-j] K[i,j]$$

Sobel gradients

Sobel combines a central finite difference in one direction with smoothing perpendicular to it. The API returns raw gradient maps and direction, plus magnitude normalized to the current image maximum.

```
Kx = [-1 0 1; -2 0 2; -1 0 1]    Ky = transpose(Kx)
Gx = I*Kx    Gy = I*Ky
Magnitude = sqrt(Gx^2 + Gy^2)    angle = atan2(Gy,Gx)
```

Binary and grayscale morphology

Represent the nonzero mask pixels by a set F and the structuring element by B . Dilation is an existential neighborhood reduction; erosion is a universal reduction.

```
F dilated by B = {z: exists b in B with z-b in F}
F eroded by B  = {z: for all b in B, z+b is in F}
Opening = erosion then dilation
Closing = dilation then erosion
```

Constant padding treats pixels outside the image as background for dilation and as foreground for erosion. Grayscale dilation is a sliding maximum padded by negative infinity.

Connected components

A raster scan assigns provisional labels using 4- or 8-neighbor adjacency. Union-find resolves label equivalences; a second pass maps roots to consecutive component IDs. This component algorithm uses sequential loops. Component properties include area, exclusive box, centroid, and optional mean colour.

3. Geometry and calibration

Bilinear interpolation

At a fractional source coordinate, use the four neighboring pixels and separable linear weights. The weights are nonnegative and sum to one; coordinates outside the image clamp to its nearest edge.

$$I(x_s, y_s) = (1-a)(1-b) I[y_0, x_0] + a(1-b) I[y_0, x_1] \\ + (1-a)b I[y_1, x_0] + ab I[y_1, x_1]$$

Rotation and perspective mapping

Rotation about the image center is an affine homogeneous transform. To avoid holes, each destination pixel is mapped backward through the inverse transform and sampled. A homography maps points up to projective scale; inverse mapping followed by division by the homogeneous coordinate creates perspective effects.

```
M = T(center) R(theta) T(-center)
R(theta) = [cos(theta), -sin(theta); sin(theta), cos(theta)]
(u,v,w) = inverse(H) * (x_destination,y_destination,1)
(x_source,y_source) = (u/w,v/w)
```

Normalized DLT

Each point correspondence contributes two homogeneous equations in the nine entries of H . Hartley normalization centers each point set and scales its mean radial distance to $\sqrt{2}$. SVD obtains the least-squares null-space vector, and the estimate is denormalized.

```
A h = 0; h is the right singular vector for the smallest singular value
H = inverse(T_destination) * H_normalized * T_source
```

RANSAC and reprojection error

RANSAC samples four correspondences per candidate, measures Euclidean reprojection errors, keeps the largest inlier set, and refits with all selected inliers. It is robust to outliers when a usable consensus exists, but does not automatically fix degenerate or poorly distributed points.

```
e_i = norm( project(H * [x_i,y_i,1]) - [xprime_i,yprime_i] )
inlier_i = (e_i < threshold)
```

4. Features and object measurements

Harris corners

Local weighted products of image gradients form a 2 by 2 structure tensor. Its eigenvalues distinguish corner-like regions, which vary in two directions, from edge-like regions.

```
M = [G_sigma * Ix^2, G_sigma * IxIy;
      G_sigma * IxIy, G_sigma * Iy^2]
R = det(M) - k * trace(M)^2
```

Hough lines

Each edge pixel votes for sampled lines in polar normal form. Accumulator peaks are line hypotheses; nearby peaks are clustered to estimate a count. Resolution and peak thresholds affect the result.

```
rho = x cos(theta) + y sin(theta)
```

Moments and shape descriptors

Foreground coordinates define area, centroid, central moments, orientation, and eccentricity. The exclusive bounding box has width $x_2 - x_1$ and height $y_2 - y_1$. The implementation reports true foreground pixel area separately from rectangle area.

```
A = count(foreground pixels)
centroid = (sum(x)/A, sum(y)/A)
phi = 0.5 atan2(2 mu11, mu20-mu02)
eccentricity = sqrt(1 - lambda2/lambda1)
```

```

extent = A/(box_width*box_height)
circularity = min(1, 4 pi A / perimeter^2)
solidity = A / convex_hull_area

```

The heuristic calls a region a circle only when circularity ≥ 0.85 and extent ≤ 0.88 ; otherwise aspect ratio separates square-like from rectangular shapes. Colour is selected as the nearest of six fixed RGB reference colours. These discrete measurements depend on mask quality and object scale.

5. Detection and non-maximum suppression

Normalized cross-correlation

For each valid image patch, NCC is Pearson correlation between the zero-mean patch and zero-mean template. It lies in $[-1,1]$ and is invariant to additive offsets and positive contrast scaling in the non-degenerate case. Flat patches or templates score zero.

$$\text{NCC}(x,y) = \text{mean}((P - \text{mean}(P)) * (T - \text{mean}(T))) / (\text{std}(P) * \text{std}(T))$$

The response map gives a score at each top-left patch location. A memory guard limits patch-statistic temporaries. Multi-scale matching searches a Gaussian pyramid and maps candidate locations back to the original scale.

Gaussian pyramid

```

G = outer([1,4,6,4,1],[1,4,6,4,1]) / 256
I[level+1] = (I[level] * G)[::2,::2]

```

IoU and NMS

Exclusive-coordinate boxes make width and height direct coordinate differences. Greedy non-maximum suppression keeps the highest-score box and discards remaining boxes whose overlap exceeds the chosen IoU threshold.

$$\text{IoU}(A,B) = \text{area}(A \text{ intersect } B) / \text{area}(A \text{ union } B)$$

$$\text{area}(A \text{ intersect } B) = \max(0, y2I - y1I) * \max(0, x2I - x1I)$$

6. Segmentation and tracking

Motion and centroid

Frame differencing widens unsigned inputs before subtraction to prevent uint8 wraparound. The motion mask marks pixels whose absolute difference meets the threshold. A binary mask centroid is the arithmetic mean of foreground coordinates; an empty mask returns None.

```

D_t[y,x] = abs(I_t[y,x] - I_(t-1)[y,x])
M_t[y,x] = 255 if D_t[y,x] >= tau, otherwise 0
centroid = (sum(x)/A, sum(y)/A)

```

Kalman centroid tracking

The filter state contains image position and velocity. A constant-velocity transition predicts the next state; position measurements update that prediction. Covariance correction uses the Joseph form and linear solves for numerical stability.

```

state = [x, y, vx, vy]^T

```

```

F(dt) = [1 0 dt 0; 0 1 0 dt; 0 0 1 0; 0 0 0 1]
state_pred = F state      P_pred = F P F^T + Q
K = P_pred H^T (H P_pred H^T + R)^-1
state = state_pred + K (measurement - H state_pred)

```

Optional Mahalanobis gating rejects implausible measurements, and confidence can adjust measurement noise. Multi-object tracking assigns detections using spatial distance and manages persistent IDs, missed frames, and track creation/deletion; it is a lightweight association method, not a global optimal solver.

HSV colour segmentation

Colour tracking thresholds hue, saturation, and value. Hue-wrap segmentation handles reds that span the angular boundary. Adaptive segmentation switches to saturation/value thresholds for near-neutral samples.

7. Neural networks and the shape CNN

Reverse-mode differentiation

The NumPy neural-network module includes convolution, pooling, dense, and ReLU layers. For a dense affine layer, reverse-mode derivatives are:

```

Z = X W + b; let delta = dL/dZ
dL/dW = X^T delta
dL/db = sum(delta over batch)
dL/dX = delta W^T
ReLU(a) = max(0,a); derivative = 1[a>0]

```

Softmax cross-entropy and SGD

```

softmax(z)_c = exp(z_c) / sum_j exp(z_j)
L = -sum_c y_c log(softmax(z)_c)
dL/dz = softmax(z) - y
parameter <- parameter - learning_rate * gradient

```

Bundled shape CNN

The trained classifier accepts a non-empty cropped binary mask and predicts Circle, Square, or Rectangle. The result includes a label, confidence, and per-class probabilities. Blank masks raise ValueError. It augments the geometric heuristic; it is not a general object detector.

Version 4.2.1 retrained with the full square rotation range and applied degradation augmentation before resizing. Bundled metadata records training configuration and a measured synthetic stress-test result; that score is not a guarantee for arbitrary real camera data.

8. Numerical practice and limitations

- Cast unsigned image arrays before signed differences or products; uint8 arithmetic can wrap.
- Use floating-point intermediates for correlations, moments, projective transforms, and covariance updates.
- Guard divisions at achromatic HSV pixels, flat NCC patches, empty masks, and projective points near infinity.

- Stride-based views avoid copying patches, but reductions and elementwise products can allocate large temporary arrays.
- Connected components, contour tracing, RANSAC, and greedy suppression include sequential steps; not all operations are loop-free.
- Results depend on encoding, thresholds, scale, mask quality, noise, and camera geometry. The package exposes algorithmic building blocks rather than automatic parameter selection.

The neural-network backward passes are covered by finite-difference gradient checks in the project tests. Other tests exercise image, tracking, detection, and calibration conventions. Python docstrings remain authoritative for exact parameter and return-value details.

9. API map and references

Main modules

core: `rgb_to_grayscale`, `threshold`. color: `histogram_equalize`, `rgb_to_hsv`. filters: `convolve2d`, `Sobel`. morphology: binary and grayscale operations, `components`. geometry: `interpolation`, `rotation`, `perspective_warp`. features: `Harris`, `Hough`, region metrics and heuristic shape classification. detection: `NCC`, `pyramids` and `NMS`. tracking: `masks` and `Kalman trackers`. calibration: `normalized DLT` and `RANSAC`. `nn/shape_cnn`: NumPy layers and bundled classifier. `streaming`: optional PyAV `StreamReader`.

References

- [1] ITU-R Recommendation BT.601-7, Studio Encoding Parameters of Digital Television, 2011. [2] C. Harris and M. Stephens, A combined corner and edge detector, Alvey Vision Conference, 1988. [3] R. Hartley and A. Zisserman, Multiple View Geometry in Computer Vision, 2nd ed., Cambridge University Press, 2004.
- [4] R. C. Gonzalez and R. E. Woods, Digital Image Processing, 4th ed., Pearson, 2018. [5] J. P. Lewis, Fast normalized cross-correlation, Vision Interface, 1995. [6] C. E. Shannon, Communication in the presence of noise, Proceedings of the IRE, 1949. [7] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning, MIT Press, 2016.

Source, tests, bundled model metadata, and release history: <https://github.com/charlie1/mycv2>